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3.8.5 Magnetic dipole moment of revolving electron

Suppose an electron undergoes circular motion around the nucleus as shown in Figure 3.41. The circulating electron in a loop is like current in a circular loop (since flow of charge is current). The magnetic dipole moment due to current carrying circular loop is

$$\vec{\mu}_L = I\,\vec{A} \qquad (3.44)$$

In magnitude,

$$\mu_L = I\,A$$

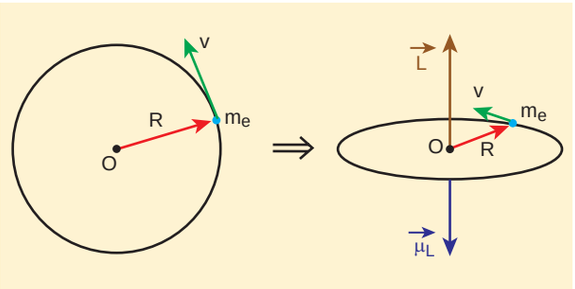


Figure 3.41 (a) Electron revolving in a circular orbit (b) Direction of magnetic dipole moment vector and orbital angular momentum vector are opposite

If T is the time period of an electron, the current due to circular motion of the electron is

$$I = \frac{-e}{T} \qquad (3.45)$$

where $-e$ is the charge of an electron. If R is the radius of the circular orbit and v is the velocity of the electron in the circular orbit, then

$$T = \frac{2\pi R}{v} \qquad (3.46)$$

Using equation (3.45) and equation (3.46) in equation (3.44), we get

$$\mu_L = -\frac{e}{\frac{2\pi R}{v}}\pi R^2 = -\frac{evR}{2} \qquad (3.47)$$

where $A = \pi R^2$ is the area of the circular loop. By definition, angular momentum of the electron about O is

$$\vec{L} = \vec{R} \times \vec{p}$$

In magnitude,

$$L = Rp = mvR \qquad (3.48)$$

Using equation (3.47) and equation (3.48), we get

$$\frac{\mu_L}{L} = -\frac{\frac{evR}{2}}{mvR} = -\frac{e}{2m} \Rightarrow \vec{\mu}_L = -\frac{e}{2m}\vec{L} \qquad (3.49)$$

The negative sign indicates that the magnetic moment and angular momentum are in opposite direction.

In magnitude,

$$\frac{\mu_L}{L} = \frac{e}{2m} = \frac{1.60\times10^{-19}}{2\times9.11\times10^{-31}} = 0.0878\times10^{12}$$

$$\frac{\mu_L}{L} = 8.78\times10^{10}\,C\,kg^{-1} = \text{ constant}$$

The ratio $\frac{\mu_L}{L}$ is a constant and also known as gyro-magnetic ratio $\left(\frac{e}{2m}\right)$. It must be noted that the gyro-magnetic ratio is a constant of proportionality which connects angular momentum of the electron and the magnetic moment of the electron.

According to Neil's Bohr quantization rule, the angular momentum of an electron moving in a stationary orbit is quantized, which means,

$$L = n\hbar = n\frac{h}{2\pi}$$

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